COMPETITIVE ANALYSIS

COMPARATIVE ANALYSIS.

75% LESS CAPEX. 45% LESS OPEX.

A detailed comparison of Tobe Energy's electrolyzer against Alkaline, PEM, Solid Oxide, and Anion Exchange systems across cost, efficiency, and longevity.

>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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Prepared: March 18, 2026 **Source:** 45VH2-GREET (Rev. Dec 2025) + Argonne emission factors **Version:** 2.0 **Classification:** Confidential

EXECUTIVE SUMMARY

Without clean power, Tobe does not qualify for 45V. With a 100% renewable PPA, each facility generates **$1.38–1.76M/year net benefit** (**$13.8–17.6M over 10 years**) after PPA costs. Tobe's **42 kWh/kg** efficiency creates a significant competitive advantage over conventional electrolyzers.

OKLAHOMA PLAINS REGION GRID ANALYSIS

SOURCE	GRID SHARE	EMISSION FACTOR	WEIGHTED CONTRIBUTION
Wind	36.6%	0 gCO2eq/kWh	0.0
Coal	31.8%	1,058 gCO2eq/kWh	336.4
Natural Gas	21.4%	487 gCO2eq/kWh	104.3
Nuclear	5.0%	2.5 gCO2eq/kWh	0.1
Hydro	4.3%	~0 gCO2eq/kWh	0.0
Biomass	0.3%	50.3 gCO2eq/kWh	0.2
Solar	0.3%	~0 gCO2eq/kWh	0.0
TOTAL	**100%**		**440.8 gCO2eq/kWh**

Source: GREET 2025, NEEDS Region "Plains"

TOBE LIFECYCLE EMISSIONS BY ELECTRICITY SOURCE

Tobe electricity consumption: ***42 kWh/kg H_2***

SCENARIO	GRID %	CLEAN %	LIFECYCLE (KGCO2E/KG)	45V CREDIT	ANNUAL VALUE*
Grid only	100%	0%	18.51	❌ None	$0
75% grid	75%	25%	13.88	❌ None	$0
50/50	50%	50%	9.26	❌ None	$0
25% grid	25%	75%	4.63	❌ None	$0
10% grid	10%	90%	1.85	**$0.75/kg**	**$675K**
5% grid	5%	95%	0.93	**$1.00/kg**	**$900K**
100% clean	0%	100%	0.00	**$3.00/kg** ⭐	**$2.7M**

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*Per facility at 900,000 kg/year nameplate production

RENEWABLE PPA ECONOMICS (100% WIND)

Cost Side

ITEM	VALUE	SOURCE
Tobe electricity per facility	**37.8 GWh/year**	42 kWh/kg × 900,000 kg
Oklahoma wind PPA rate	**$0.025–0.035/kWh**	Market rates, 2025–2026
Annual PPA cost	**$945K–$1.3M/year**	37.8 GWh × rate
vs. grid electricity ($0.035/kWh)	$1.32M/year	Current Oklahoma industrial
Incremental PPA cost	$0 – marginal	Wind PPA may be at or below grid

Revenue Side (45V PTC)

ITEM	VALUE
Credit rate	**$3.00/kg**
Annual production	**900,000 kg**
Annual PTC revenue	**$2.7M**
Credit window	10 years
Lifetime PTC per facility	**$27M**

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NET BENEFIT ANALYSIS

SCENARIO	ANNUAL PPA COST	ANNUAL PTC	NET BENEFIT	10-YEAR NET
PPA @ $0.025/kWh	$945K	$2.7M	**$1.76M/year**	**$17.6M**
PPA @ $0.030/kWh	$1.13M	$2.7M	**$1.57M/year**	**$15.7M**
PPA @ $0.035/kWh	$1.32M	$2.7M	**$1.38M/year**	**$13.8M**

MULTI-FACILITY IMPACT (FINANCIAL MODEL ALIGNMENT)

FACILITIES	ANNUAL PTC (TOTAL)	10-YEAR PTC (TOTAL)
1 (FY1–2)	**$2.7M**	**$27M**
3 (FY3)	**$8.1M**	**$81M**
4 (FY4+)	**$10.8M**	**$108M**
12 (FY7)	**$32.4M**	**$324M**

Financial model v8.5.3 assumes 4 facilities qualify (begin construction by Dec 2027)

TOBE'S EFFICIENCY ADVANTAGE

ELECTROLYZER	KWH/KG	MWH/YEAR (900K KG)	PPA COST @$0.03	NET AFTER PTC
Tobe	**42**	37,800	$1.13M	**$1.57M**
Typical PEM	55	49,500	$1.49M	$1.21M
Typical ALK	60	54,000	$1.62M	$1.08M

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Tobe's 42 kWh/kg efficiency means 24–30% less clean power needed than competitors to hit the same emissions threshold. Lower electricity bill + same PTC = wider margin.

KEY RISKS

1. **45V begin-construction deadline**: May be accelerated to 2025 under potential legislation. Must break ground early to qualify.
2. **Hourly matching requirements**: Treasury may require temporal matching (not just annual RECs). Hourly-matched PPAs cost more.
3. **Additionality**: 45V may require NEW renewable capacity, not existing wind farms.
4. **Regulatory change**: Congress could modify/repeal. Model with and without.

RECOMMENDATION

Sign a 100% renewable wind PPA for every facility. The math is unambiguous:

- Even at the highest PPA rate ($0.035), net benefit is **$1.38M/year per facility**
- Oklahoma wind is abundant and cheap
- Tobe's efficiency amplifies the advantage vs. competitors
- The PTC alone could fund **$13.8–17.6M per facility** over 10 years

NEXT STEPS:

1. Identify Oklahoma wind PPA providers (NextEra, AES, Invenergy)
2. Get indicative pricing for 10-year, 37.8 GWh/year contract
3. Confirm Treasury hourly matching requirements for 45V
4. Begin construction documentation before deadline

Analysis using 45VH2-GREET (Rev. December 2025) emission factors, Argonne National Laboratory GREET model, Plains region, NEEDS electricity distribution

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